CardDemo: COBOL → Rust

Project progress report: sessions 1 and 2

Prepared with Claude Code (Opus 5.5)
2026-11-06

Status at a glance

Where we are

Session 1

- Requirements specification (PDF)
- Rust workspace, 6 crates, **all 30 base programs** (17 CICS online + terminal UI, 13 batch + JCL runner)
- Manual test suite (PDF, 100% FR-group coverage)
- Live-demo script, CLI, one-command reset

Session 2

- **Differential test vs. the original COBOL** (GnuCOBOL): 13 batch programs, **all outputs identical**
- **Manual test suite executed: 41 / 41 pass**
- Clippy clean; multi-terminal tests

Found and fixed in session 2

- 9 fidelity gaps in the Rust port, all invisible to the port's own tests (details on the next slides)
- 1 new defect in the original: KD-12
- 8 test-case texts corrected against the COBOL

Open

- Optional modules (IMS/DB2/MQ): out of scope by decision
- Live demo rehearsal

Scope and approach

Scope (agreed 2026-11-05)

In scope	Out of scope
Base app: COBOL, CICS, VSAM, JCL, assembler	Authorisations (IMS / DB2 / MQ)
20.6k lines COBOL, 2.8k copybook, 4.5k BMS	Transaction types (DB2)
Terminal UI (3270 style), SQLite, fixed-width I/O	MQ account and date inquiry

Decisions: faithful translation (same messages, layouts, bytes, **and defects**); GnuCOBOL for batch differential tests; live demo planned.

Architecture

Crate	Replaces	L
`cobol`	COBOL data semantics: zoned, COMP-3, edit pictures, cp037 + generated copybooks	1
`vsam`	VSAM KSDS/AIX on SQLite: EBCDIC key order, CICS browse rules	
`cics`	BMS maps parsed from the original sources; XCTL / RETURN, file control, TDQ	
`online`	the 17 CICS programs, one function per COBOL paragraph	4
`batch`	13 batch programs + JCL jobs (SORT, REPRO, GDG, internal reader)	1
`carddemo`	binary: ratatui TUI, `job`, `batch`, `reset`, `script`	

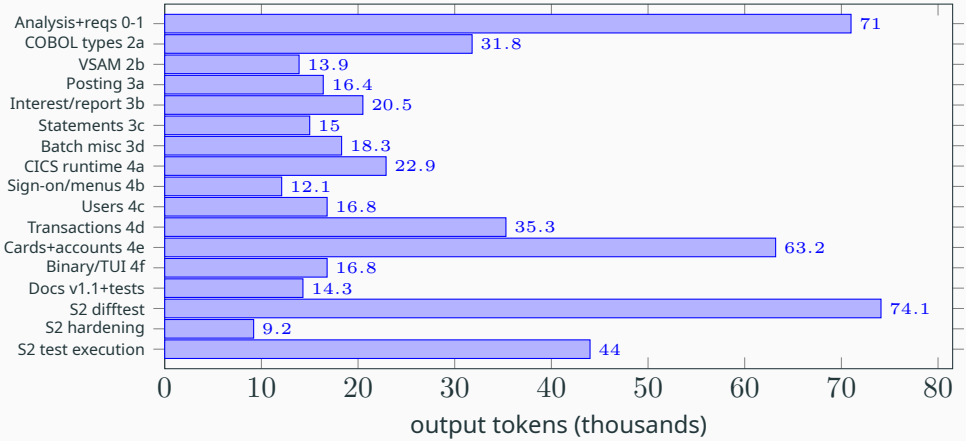
About **9.7k lines of Rust** (plus 1.1k generated) replace about **27k lines** of COBOL, copybooks and BMS.

Plan versus actual

Phases

#	Phase	Plan	Actual
0	Inventory	call graph, CRUD matrix	done, scripted extraction
1	Requirements PDF	FR / NFR / traceability	26 pp, 165 FR, 12 NFR, 10 KD
2	Foundation crates	types, records, store	done; records generated from copybooks
3	Batch	13 programs + jobs	done; export/import round trip byte-identical
4	Online + TUI	17 programs	done; scripted terminal journeys pass
5	Optional modules	—	dropped (scope decision)
6	Manual test PDF	per requirement	40 cases, 122 steps
7	Hardening, demo	runbook	demo script, README; GnuCOBOL diff test (13 p
6b	Test execution	—	41 / 41 pass (scripted terminal + difftest)
8	Progress deck	this document	updated after session 2

Effort by phase (output tokens)



Time and tokens (from the Claude Code transcripts)

Metric	Session 1	Session 2	Total
Active time (gaps > 10 min excluded)	1.1 h	0.5 h	1.6 h
Assistant turns / tool calls	194 / 212	180 / 187	374 / 399
Output tokens	367k	137k	504k
Prompt cache reads	90.9M	41.3M	132.2M
Prompt cache writes	0.76M	0.32M	1.08M
Commits on branch `rust-port`	15	3	18

Measured with `carddemo-rs/tools/usage.py` and `tools/phase_metrics.py`. Cache reads dominate: every turn re-reads the conversation.

Quality and fidelity

Testing

- **Unit tests:** numerics (zoned/COMP-3/binary), edit pictures, NUMVAL, cp037, VSAM browse rules, BMS parser, CEEDAYS dates
- **Record tests:** generated layouts decode the original EBCDIC sample data
- **Batch:** the full nightly cycle (TRANBKP → POSTTRAN → INTCALC → COMBTRAN → TRANREPT → CREASTMT); CBEXPORT → CBIMPORT is byte-identical
- **Online:** scripted 3270 journeys through all 17 programs (sign-on, admin, lists, updates, payments, report submission)
- **Multi-terminal:** a concurrent update is detected; parallel payments get unique transaction ids
- **Manual suite:** executed as one scripted test per case: **41 / 41 pass**
- **Result:** all tests pass (`cargo test --workspace`), clippy clean

Differential test against the original COBOL

Job / program	Outputs compared
POSTTRAN / CBTRN02C	TRANSACT, ACCTDATA, TCATBALF, DALYREJS, RC, SYSOUT
INTCALC / CBACT04C	ACCTDATA, SYSTRAN, RC, SYSOUT
TRANREPT / CBTRN03C	report (519 lines), RC, SYSOUT
CREASTMT / CBSTM03A+B	statement text + HTML (7.9k lines), RC
READACCT/CARD/XREF/CUST, DALYVRFY	OUTFILE (COMP-3), ARRYFILE, VB file, SYSOUT
CBEXPORT / CBIMPORT	export file, 6 import outputs (incl. error path), SYSOUT

The original sources were compiled unchanged with GnuCOBOL 3.2, except for three patches: CBSTM03A's z/OS control-block walk is stubbed out, the KD-12 FD is fixed, and tabs are expanded. Both sides use the same input snapshots and a fixed clock. The documented platform differences (DISPLAY sign style, INITIALIZE sign) are normalised. `python3 carddemo-rs/difftest/run.py`

What the comparison and the test run found

Area	Rust port before	Original behaviour (now matched)
Storage without VALUE	spaces	binary zeros: FILLER bytes of posted/interest tra
INITIALIZE	blanked FILLER	leaves FILLER and REDEFINES alone
CBSTM03A HTML	`DELIMITED BY ' '`	`DELIMITED BY '  '`: full names and addresses
CBCUS01C, CBACT01C	one DISPLAY per record	a second DISPLAY in the main loop
CBIMPORT	no “Errors Written” line	statistics line present
COTRN00C search	key = spaces → NOTFND	key = X’00’ → finds the init record
Account messages	1 space, cut at 78	`file.  Resp:`, cut at 75 (WS-RETURN-MSG)

Defects found in the original (kept on purpose)

ID	Where	Effect
KD-01	INTCALC	the last account's interest is never written back
KD-02/03	TRANREPT	last amount counted twice; no final account total
KD-07	card update	the CVV is reset to 000
KD-08	account update	the "blank phone" test checks the area code twice
KD-10	user/tran list	search past the last key → AEIP abend
KD-11	user/tran list	PF7 then PF8 on page 1 → empty page
KD-12	CBEXPORT/CBIMPORT	RECORD KEY outside the file's record: does not compile as shipp

Also: the COACTVW map blanks most of the `Tran:` header, and the sample data fails the COACTUPC state/zip check for 48 of 50 customers.

Lessons learned

- **Read the source, never a filtered view.** A space-collapsing reader and a grep filter both produced wrong requirements (caught and fixed: KD-06 was withdrawn).
- **COBOL semantics hide in ordering.** For example `SET NEXT-PAGE-NO` *before* the commarea MOVE.
- **The 3270 matters:** FSET fields, nulls versus spaces, overlapping map fields.
- **Generate from the originals:** copybooks → Rust records, BMS → screens, CSLKPCDY → lookups.
- **Run the original.** Unit and journey tests check what we *think* the code does. The GnuCOBOL comparison found 6 kinds of gap at once, and executing the suite found 3 more. Most were about storage nobody writes: FILLERs and fields without VALUE.
- **Execute the test suite, don't just write it:** 8 of 41 case texts had wrong expectations.

Next steps

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1. Live demo rehearsal (`docs/DEMO.md`); `carddemo-rs/difftest/run.py` can be part of the demo
2. Optional: an online differential test (needs a CICS emulator; out of scope so far)
3. Optional modules (IMS / DB2 / MQ) if the scope is extended

Demo

```
carddemo reset          # 5 ms
carddemo online         # sign on as USER0001 / PASSWORD
carddemo batch          # nightly cycle in about 20 ms
```

Branch `rust-port`: `docs/requirements.pdf`, `docs/manual-test-suite.pdf`, `docs/DEMO.md`.